MILITARY SPECIFICATION

CLOTH, FLANNEL, COTTON (FLAME RETARDANT TREATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of flannel cotton cloth which has been treated with a flame retardant finish.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging, Packing of Cotton and Cotton Synthetic Fiber Blend Fabrics (excluding duck fabrics)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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MILITARY

MIL-T-43548 - Thread, Polyester, Core: Cotton-, Rayon-, or Polyester Covered

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-4 - Glossary of Fabric Imperfections

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspections by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained from the Federal Trade Commission, Washington, DC 20580).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 4.3 and 6.2), the contractor shall furnish a sample for first article inspection and approval.

3.2 Material.

3.2.1 Yarn. The yarn shall be made from cotton which has been carded, drawn, and spun into singles yarn for both the warp and the filling, when tested as specified in 4.5.

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3.3 Weave. The weave shall be a 3 or 4 harness twill, when tested as specified in 4.5.

3.4 Color. The cloth shall be unbleached.

3.5 Physical requirements. The finished cloth shall conform to the physical requirements shown in Table I when tested as specified in 4.5.

TABLE I. Physical requirements

Characteristics	Minimum	Maximum
Weight, oz/yd^2 (gm/m^2)		
Greige	8.00 (271.2)	-
Finished	8.75 (296.6)	9.75 (330.5)
Tear Strength, lbs (N)		
Warp	5.00 (22.2)	-
Filling	4.50 (20.0)	-
Stiffness, in-lbs (N.m)		
Warp	-	.006 (.0007)
Filling	-	.004 (.0004)
Air permeability, ft^3/min/ft^2 (cm^3/sec/cm^2)	20 (10.2)	-
Yarns per inch (2.54 cm)		
Warp	62	-
Filling	56	-
Breaking strength, lbs (N)		
Warp	65 (289.2)	-
Filling	40 (178.0)	-
Flame resistance		
Initially and after 25 launderings:		
After flame time, seconds	-	2
After glow time, seconds	-	5
Char length, inches (2.54 cm)	-	5

3.6 Finish. The cloth shall be napped on the back side, scoured, desized, and given an approved flame retardant treatment (see 6.4 and 6.5).

3.6.1. Toxicity approval. Only those flame retardant treatments already evaluated and approved by the appropriate medical service in regards to toxicity and solicited in the invitation for bids or requests for proposals shall be considered acceptable for the related procurement (see 6.4).

3.7 Nonfibrous material. Prior to the application of the flame retardant treatment, the cloth shall contain no more than 3.0 percent starch and protein content including chloroform-soluble and water-soluble material, when tested as specified in 4.5.

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3.8 Width. The width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width, inclusive of selvages when fly shuttle looms or shuttleless with tuck-in selvage looms are used. For all other shuttleless looms the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe.

3.9 Seam efficiency. The cloth shall have a seam efficiency of not less than 85 percent when tested as specified in 4.5.

3.10 Dimensional stability. The finished cloth shall not shrink nor elongate more than 2.0 percent in either the warp or filling direction when tested as specified in 4.5. The preshrinking process used shall not be identified by name or by trademark, either on the cloth, ticket, or package.

3.11 pH. The pH value of the water extract of the finished cloth shall be [illegible] when tested as specified in 4.5.

3.12 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up in full width rolls as specified in 5.1.

3.13 Fiber identification. Each piece shall be labeled, or ticketed [illegible] fiber content in accordance with the Textile Fiber Products Identification Act.

3.14 Face identification. The face side of the cloth shall be identified by stamping on that side with the word "Face" at each end of the roll.

3.15 Workmanship. The cloth shall conform to the quality and grade of product established by this document. The occurence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 3.1 and 6.2), the first article shall be visually examined for appearance, color, and finish, and shall be tested for physical and chemical properties in accordance with the methods specified in 4.5.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.3.

4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the napped side only. When the total yardage in the roll does not exceed 100 yards (91.4m) the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m), only 100 yards (91.4m) shall be examined. All defects as defined in Section 1 of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) (0.9m) shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only the slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.3), "D" for slubs, or 3-1/2 as applicable, and "C" for knots shall be scored. No linear yard (increments of 1 yard (0.91m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 thru 10,000	13
10,001 and over	20

1/ Only one roll shall be taken from any shipping container, unless the number of shipping containers in the lot are less than the number of rolls required for sampling, in which case, all shipping containers shall be sampled.

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The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined exceeds 28 points. The lot shall be unacceptable if the individual points per 100 square yards ($83.6m^2$) of two or more individual rolls exceeds 40 points. If one roll exceeds the point level per 100 square yards ($83.6m^2$), a second sample, the same size as the final sample, shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 38 points per 100 square yards ($83.6m^2$). Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth (inches) x total yards inspected}} = \text{Points per 100 square yards } (83.6m^2)$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3 inches (7.6cm) in any dimension	= one point
For defects exceeding 3 inches (7.6cm) but not exceeding 6 inches (15.2cm) in any dimension	= two points
For defects exceeding 6 inches (15.2cm) but not exceeding 9 inches (22.8cm) in any dimension	= three points
For defects exceeding 9 inches (22.8cm) in any dimension	= four points

The following defects, when present, shall be scored 4 points for each yard (0.91m) in which they occur:

Baggy, ridgy, or wavy cloth
Width less than specified
Excessive neppiness
Objectionable odor

4.4.2.2 Examination for length.

4.4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards (1.8m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

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4.4.2.3 Examination for face identification, non-identification of preshrinking process, and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Face identification missing from either or both ends
Face identification on wrong side
Preshrinkage process identified by name or trademark on cloth, ticket, or package
Not labeled or ticketed in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act

4.4.3 Examination of packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine whether packaging, packing, and marking comply with section 5 requirements.

4.5 Tests. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 5 continuous yards (4.6m) full width of finished cloth, and 1/4 yard (.23m) full width of cloth prior to the application of the finish for determination of nonfibrous materials and greige weight. The lot size shall be expressed in units of 1 yard (0.9m) each. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size shall be as shown in the list below.

Lot size	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. Test methods

Characteristics	Requirement paragraph	Test method
Cotton identification	3.2.1	1200 1/
Yarn ply	3.2.1	1/
Weave	3.3	visual 2/
Color	3.4	visual
Weight	3.5	5041
Tear strength	3.5	5132
Stiffness	3.5	5202
Air permeability	3.5	5450
Yarns per inch	3.5	5050
Breaking strength	3.5	5100
Flame resistance		
Initially	3.5	5903
After 25 launderings	3.5	5556, 5903
Finish		
Napped	3.6	visual
Desized	3.6	1/
Scoured	3.6	1/
Nonfibrous	3.7	2611
Seam efficiency	3.9	5110 3/
Dimensional stability	3.10	5556
pH	3.11	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ One determination shall be made on each sample unit and the result reported as "pass" or "fail".

3/ The thread shall be polyester core, cotton or rayon covered, size 50, 2 or 3 ply for the needle thread, and size 70, 2 or 3 ply for the looper thread of MIL-T-43548. The needle size shall be 0.040 (± 0.001) inch.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation packaging shall be level A or Commercial as specified (see 6.2).

5.1.1. Level A or Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A, B, and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

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6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in flame resistant anti-flash gloves.

6.2 Ordering data. Procurement documents should specify the following.

a. Title, number, and date of this document.
b. When first article sample is required (see 3.1 and and 4.3). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, quantity, testing and approval of the first article.
c. Width required (see 3.8).
d. Length, if other than specified (3.12).
e. Levels of packaging and packing required (see 5.1 and 5.2).

6.3 Sears Fabric Defect Scales are available from Sears Roebuck and Company, Department 817 (Attn: BSC 23-29), Sears Tower, Chicago, IL 60684 (see 4.4.2.1).

6.4 Only those treatments listed in the invitation for bids or request for proposal shall be considered acceptable for related procurement. Information on currently approved material may be obtained from Defense Personnel Support Center, Philadelphia, PA 19101 (see 3.6).

6.5 A cotton fire retardant treatment (FRT) with a precondensate of a tetrakis-hydroxymethyl phosphonium salt cured in a gaseous ammonia atmosphere (precondensate/ammonia cure) has been found to meet the document requirements (see 3.6).

6.6 Subject term (key word) listing.

Cloth, flannel
Fiber, cotton
Finish, flame retardant
Gloves, anti-flash, flame resistant

Custodian:
Navy - NU
Army - GL
Air Force - 99

Preparing Activity:
Navy - NU

Review Activity:
Army - MD
Air Force - 82
DLA - CT

Project No. 8305-0156

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MII-C-24933	CLOTH, FLANNEL, COTTON (FLAME RETARDANT TREATED)

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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